Equilibria, Stability and Hopf Bifurcation

Rosenzweig–MacArthur model for lynx and hare

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1 Equilibria, Stability and Hopf Bifurcation

Throughout this section we analyse the dimensional Rosenzweig–MacArthur model

$$\frac{dH}{dt} = f(H,P) = rH\Big(1-\frac{H}{K}\Big) - \frac{aHP}{b+H}, \qquad \frac{dP}{dt} = g(H,P) = \frac{eaHP}{b+H} - mP, \tag{1}$$

with all six parameters r, K, a, b, e, m strictly positive, as defined in the model section. Every stability statement below is tied either to a derivation given here, to a cited theorem, or to a numerical check; Table 3 at the end of the section records which is which.

1.1 Well-posedness: invariance and boundedness

Lemma 1. *The closed quadrant $\{H \geq 0,\, P \geq 0\}$, both coordinate axes, and the open quadrant are invariant under* (1)*. Every solution starting in the quadrant is bounded, with*

$$\limsup_{t\to\infty} H(t) \leq K, \qquad \limsup_{t\to\infty}\big(eH(t)+P(t)\big) \leq \frac{eK(r+m)^2}{4rm}.$$

Proof. Both right-hand sides factor as $H\cdot(\cdot)$ and $P\cdot(\cdot)$, so $H=0$ and $P=0$ are invariant lines and, by uniqueness of solutions, no orbit can cross them. Since $\dot H \leq rH(1-H/K)$, comparison with the logistic equation gives $\limsup H \leq K$. For $W = eH + P$ the predation terms cancel: $\dot W = erH(1-H/K) - mP = eH\big(r(1-H/K)+m\big) - mW$. The quadratic $H\big(r(1-H/K)+m\big)$ is maximised at $H = K(r+m)/(2r)$ with value $K(r+m)^2/(4r)$, so $\dot W \leq eK(r+m)^2/(4r) - mW$ and the bound follows. □

Boundedness is what later allows the Poincaré–Bendixson theorem to be used.

1.2 Equilibria and biological feasibility

Setting $f = g = 0$:

- $P = 0$ in $g = 0$ leaves $rH(1-H/K) = 0$, giving the *origin* $E_0 = (0,0)$ and the *prey-only* equilibrium $E_1 = (K, 0)$.
- $P \neq 0$ requires $eaH/(b+H) = m$, i.e.

$$H^* = \frac{mb}{ea-m}, \tag{2}$$

and then $f = 0$ with $H = H^* \neq 0$ gives

$$P^* = \frac{r}{a}\Big(1-\frac{H^*}{K}\Big)(b+H^*) = \frac{ebr\big(K(ea-m)-bm\big)}{K(ea-m)^2}. \tag{3}$$

This is the *coexistence* equilibrium $E^* = (H^*, P^*)$.

Proposition 1 (Feasibility). *E_0 and E_1 always lie in the closed quadrant. E^* lies in the open quadrant if and only if*

$$ea > m \quad \textit{and} \quad H^* < K \qquad \Longleftrightarrow \qquad m < \frac{eaK}{b+K}.$$

Proof. $H^* > 0$ iff $ea > m$. Given this, $P^* > 0$ iff $1 - H^*/K > 0$. Rearranging $mb/(ea-m) < K$ gives $m(b+K) < eaK$. □

Biologically, ea is the predator's maximal per-capita birth rate (attained when prey is abundant), so $ea > m$ says lynx can outgrow their mortality at all; $H^* < K$ says the hare level lynx need to break even is actually reachable by the prey on its own. Note that H^* does not depend on r or K: the prey level at coexistence is set entirely by the predator.

1.3 Jacobian

Differentiating (1),

$$J(H,P) = \begin{pmatrix} r\Big(1 - \dfrac{2H}{K}\Big) - \dfrac{abP}{(b+H)^2} & -\dfrac{aH}{b+H} \\[2ex] \dfrac{eabP}{(b+H)^2} & \dfrac{eaH}{b+H} - m \end{pmatrix}. \tag{4}$$

For a planar system, with $\tau = \operatorname{tr} J$ and $\Delta = \det J$, an equilibrium is a saddle if $\Delta < 0$; is asymptotically stable if $\Delta > 0$, $\tau < 0$; and is unstable if $\Delta > 0$, $\tau > 0$. It is a focus when $\tau^2 < 4\Delta$ and a node when $\tau^2 > 4\Delta$.

1.4 Local stability of each equilibrium

Origin. $J(E_0) = \operatorname{diag}(r, -m)$, with eigenvalues $r > 0$ and $-m < 0$. E_0 is *always a saddle*: the P-axis is its stable manifold (lynx alone die out) and the H-axis its unstable manifold (hares alone grow). Joint extinction is never an attractor.

Prey-only equilibrium.

$$J(E_1) = \begin{pmatrix} -r & -\dfrac{aK}{b+K} \\[2ex] 0 & \dfrac{eaK}{b+K} - m \end{pmatrix}, \qquad \lambda_1 = -r, \quad \lambda_2 = \frac{eaK}{b+K} - m.$$

E_1 is a stable node when $\lambda_2 < 0$, i.e. exactly when E^* is *not* feasible, and a saddle when $\lambda_2 > 0$, i.e. exactly when E^* is feasible. At $K = H^*$ the two equilibria coincide and exchange stability (a transcritical bifurcation). λ_2 is the invasion rate of a rare lynx population into a hare population at carrying capacity.

Coexistence equilibrium. At E^* the predator equation gives $eaH^*/(b+H^*) = m$, so $J_{22} = 0$. Using the prey nullcline, $aP^*/(b+H^*) = r(1 - H^*/K)$, to eliminate P^* from J_{11}:

$$J_{11} = r\Big(1 - \frac{2H^*}{K}\Big) - \frac{b}{b+H^*}\, r\Big(1 - \frac{H^*}{K}\Big) = \frac{rH^*\,(K - b - 2H^*)}{K\,(b+H^*)}.$$

Hence

$$\tau(E^*) = \frac{rH^*}{K(b+H^*)}\,\big(K - K_H\big), \qquad K_H := b + 2H^* = \frac{b\,(ea+m)}{ea-m}, \tag{5}$$

$$\Delta(E^*) = -J_{12}J_{21} = \frac{aH^*}{b+H^*} \cdot \frac{eabP^*}{(b+H^*)^2} = \frac{rmb\,(K - H^*)}{K(b+H^*)}. \tag{6}$$

Theorem 1 (Local stability of E^*)**.** *Suppose E^* is feasible ($H^* < K$). Then $\Delta(E^*) > 0$, so E^* is never a saddle, and*

$$E^* \text{ is locally asymptotically stable if } K < K_H, \qquad E^* \text{ is unstable (a repeller) if } K > K_H.$$

Proof. $\Delta > 0$ follows from (6) since $K > H^*$. The sign of τ is the sign of $K - K_H$ by (5). Apply the trace–determinant classification. □

Remark 1. The intrinsic hare growth rate r appears in τ and Δ only as a positive prefactor. It therefore sets how fast trajectories spiral, but has *no effect* on whether E^* is stable. Near feasibility (K just above H^*), $\Delta \to 0$ while $\tau < 0$, so E^* is a stable node; as K increases it becomes a stable focus once $\tau^2 < 4\Delta$ (at $K \approx 42.5$ for the illustrative parameters of §1.8), so convergence is oscillatory well before the Hopf point.

Table 1: Equilibria of (1) and their local stability.

Equilibrium	Feasible when	Eigenvalues / invariants	Type
$E_0 = (0, 0)$	always	$r,\ -m$	saddle
$E_1 = (K, 0)$	always	$-r,\ \frac{eaK}{b+K} - m$	stable node if $K < H^*$ or $ea \le m$; saddle if $K > H^*$
$E^* = (H^*, P^*)$	$ea > m,\ H^* < K$	$\tau \propto K - K_H,\ \Delta > 0$	stable if $K < K_H$; unstable if $K > K_H$

1.5 Nullclines and the phase plane

Prey nullclines ($\dot{H} = 0$): the P-axis $H = 0$, and the parabola

$$P = \phi(H) := \frac{r}{a}\Big(1 - \frac{H}{K}\Big)(b + H), \tag{7}$$

which is concave, vanishes at $H = -b$ and $H = K$, cuts the P-axis at rb/a, and peaks at

$$H_{\text{peak}} = \frac{K - b}{2}, \qquad \phi(H_{\text{peak}}) = \frac{r(K + b)^2}{4aK}.$$

$\dot{H} > 0$ below the parabola and $\dot{H} < 0$ above it (for $H > 0$).

Predator nullclines ($\dot{P} = 0$): the H-axis $P = 0$, and the vertical line $H = H^*$. $\dot{P} > 0$ to the right of $H = H^*$ (enough prey) and $\dot{P} < 0$ to the left.

Flow. The two interior nullclines divide the open quadrant into four regions. Starting below the parabola to the right of H^*, both species increase; crossing the parabola hares start to fall while lynx still rise; crossing $H = H^*$ both fall; crossing back below the parabola hares recover while lynx are still falling. The flow therefore circulates *counter-clockwise* around E^* in the (H, P) plane, and lynx peaks lag hare peaks. Near the Hopf point the lag is a quarter-period; it grows with K (numerically, 2.5 to 4.5 years for regimes D–F below).

Graphical stability criterion. Since $K_H = b + 2H^*$ is equivalent to $H^* = (K - b)/2 = H_{\text{peak}}$, Theorem 1 says:

E^* is stable if the predator nullcline cuts the prey parabola to the right of its hump, and unstable if it cuts to the left.

This is the original Rosenzweig–MacArthur graphical criterion [5]. It also gives the sign of J_{11} directly: $J_{11} = a\,H^*\phi'(H^*)/(b + H^*)$, so a rising nullcline (left of the hump) means local prey self-amplification ($J_{11} > 0$). Figure 1 shows both cases.

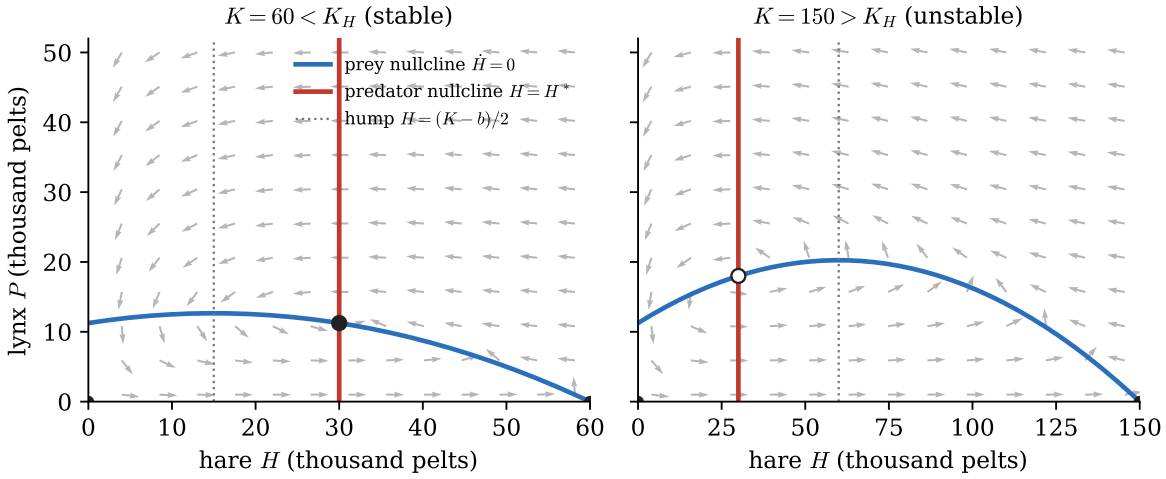


Figure 1: Nullclines and direction field for the illustrative parameters ($r = 1.5$, $a = 4$, $b = 30$, $e = 0.4$, $m = 0.8$; $H^* = 30$, $K_H = 90$). Left: $K = 60$; the predator nullcline meets the parabola right of its hump and E^* (filled) is a stable focus. Right: $K = 150$; the intersection is left of the hump and E^* (open) is unstable. Small dots mark E_0 and E_1.

1.6 Hopf bifurcation with K as the parameter

We fix r, a, b, e, m with $ea > m$ and vary K over (H^*, ∞). Note H^* is independent of K, while P^* increases with K.

Theorem 2 (Hopf bifurcation). *At $K = K_H = b + 2H^*$ the coexistence equilibrium undergoes a Hopf bifurcation:*

(i) $\tau(E^*) = 0$ *and* $\Delta(E^*) = \dfrac{rmb\,(K_H - H^*)}{K_H(b + H^*)} = \dfrac{rmb}{K_H} > 0$, *so $J(E^*)$ has eigenvalues $\pm i\omega_0$ with*

$$\omega_0 = \sqrt{\frac{rmb}{b + 2H^*}}, \qquad T_0 = \frac{2\pi}{\omega_0};$$

(ii) (transversality) the real part of the eigenvalues, $\tau/2$, crosses zero with nonzero speed:

$$\frac{d}{dK}\left(\frac{\tau}{2}\right)\Big|_{K=K_H} = \frac{rH^*}{2(b + H^*)(b + 2H^*)} > 0.$$

Proof. (i) Substitute $K = K_H$ into (5) and (6); $K_H - H^* = b + H^*$ simplifies Δ. (ii) Differentiate (5) in K: $\frac{d}{dK}\frac{K - K_H}{K} = \frac{K_H}{K^2}$, which equals $1/K_H$ at $K = K_H$. All symbolic steps were checked with SymPy (`scripts/joel_symbolic_derivation.py`) and the closed forms are tested against the shared model code (`tests/test_stability.py`). □

Conditions (i)–(ii) establish that a pair of complex eigenvalues crosses the imaginary axis. To conclude that a small periodic orbit is born, and on which side, the Hopf theorem also needs the first Lyapunov coefficient ℓ_1 to be nonzero [3]: $\ell_1 < 0$ gives a *supercritical* bifurcation (a stable cycle for $K > K_H$) and $\ell_1 > 0$ a subcritical one.

What is analytic and what is numerical. We separate four statements.

1. **Analytic (derived here):** the threshold K_H, conditions (i)–(ii), and the exchange of stability at E^*.

2. **Analytic (derived here): existence of a periodic orbit for $K > K_H$.** Take any initial condition in the open quadrant. By Lemma 1 its ω-limit set Ω is nonempty and compact. Ω cannot contain E^*, which is a repeller. Ω cannot be $\{E_1\}$, because the stable manifold of the saddle E_1 is the H-axis. If Ω contained E_1 together with other points, it would contain a point of E_1's stable manifold on the H-axis and hence the whole axis orbit through it, which tends backward to E_0; repeating the argument at E_0 would put a point of the P-axis in Ω, whose backward orbit is unbounded, contradicting compactness. So Ω contains no equilibria, and the Poincaré–Bendixson theorem [6] makes Ω a periodic orbit.

3. **Cited theorems:** Cheng [1] proved the limit cycle is *unique* for $K > K_H$; since every interior orbit other than E^* tends to a periodic orbit, that unique cycle attracts them all and is orbitally stable. Hsu [2] proved that a locally stable E^* is *globally* stable in the open quadrant whenever the prey nullcline is concave, which our parabola (7) is, so E^* attracts every interior orbit for $H^* < K < K_H$. We use these results without reproducing their proofs.

4. **Numerical (for our parameter set only):** evaluating Kuznetsov's formula for ℓ_1 at $K = K_H$ gives $\ell_1 \approx -4.0 \times 10^{-4} < 0$ (and -1.3×10^{-3} for a second set $r = 0.7$, $m = 0.5$, $b = 20$), so the bifurcation is supercritical, consistent with the cited uniqueness result. Simulations confirm the supercritical signature: the post-transient cycle half-amplitude in H scales as $(K - K_H)^{1/2}$ (fitted log–log slope 0.507 for $K - K_H \in [0.5, 8]$), and the period tends to T_0.

None of this says that the *real* hare–lynx data sit on a limit cycle. That depends on whether fitted parameters satisfy $K > K_H$, and with about two oscillations in 1900–1920 a slowly damped oscillation (K slightly below K_H) can fit the same data. That question belongs to the fitting sections and must be answered there with uncertainty.

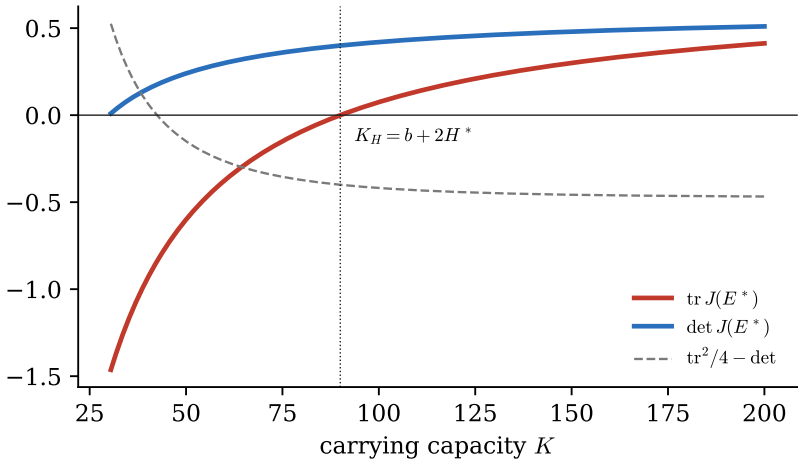


Figure 2: Trace and determinant of $J(E^*)$ as functions of K (illustrative parameters). The determinant stays positive for all feasible $K > H^* = 30$, while the trace changes sign at $K_H = 90$. The dashed curve is negative (focus) for $K > 42.5$.

1.7 The paradox of enrichment

Increasing K (enriching the hares' environment) leaves H^* unchanged and only raises P^*: every extra unit of carrying capacity is passed to the predator. Graphically, the prey parabola's hump

$(K-b)/2$ moves right while the predator nullcline stays at H^*, so for large enough K the intersection falls left of the hump and the equilibrium destabilises. Past K_H the system settles on a limit cycle whose amplitude grows with K. For our illustrative parameters, at $K = 150$ the hare minimum on the cycle is about 1.1 thousand, compared with $H^* = 30$. In a finite, stochastic population such deep troughs bring a real risk of extinction. Enrichment thus *destabilises* the system, the paradox named by Rosenzweig [4].

The mechanism is the saturating Holling type II response. At low H each lynx's consumption rises almost linearly with prey, which damps fluctuations; near and above b it saturates, so per-capita predation risk $aP/(b+H)$ *falls* as hares increase. That positive feedback on prey growth is exactly the term making $J_{11} > 0$ left of the hump.

Link to Lotka–Volterra. In the classical model ($K = \infty$, linear response) the analogous Jacobian at coexistence has $\tau \equiv 0$ for every parameter value, so it sits permanently on the Hopf boundary. That is why its neutral cycles are structurally unstable: any small change in the model's form moves it to one side or the other. Rosenzweig–MacArthur makes the side explicit through the sign of $K - K_H$.

1.8 Illustrative regimes and numerical validation targets

These parameters are *illustrative, not fitted*: H and P in thousands of pelts, t in years, with $r = 1.5,\ a = 4,\ b = 30,\ e = 0.4,\ m = 0.8$. Then $ea = 1.6 = 2m$, so $H^* = b = 30$, $K_H = 90$, $\omega_0 = \sqrt{0.4} \approx 0.632$ and $T_0 \approx 9.94$ years, close to the observed decadal cycle. Table 2 gives the expected behaviour on each side of the threshold; the simulation section reproduces these with RK45 (`rtol`$= 10^{-10}$, `atol`$= 10^{-12}$, 3000-year burn-in, start $(20, 5)$).

Table 2: Regimes in K (other parameters fixed). Eigenvalues are of $J(E^*)$, except in A where they are of $J(E_1)$.

	K	Linear prediction	Eigenvalues	Long-run behaviour (simulated)
A	25	E^* infeasible ($K < H^*$)	$-1.5,\ -0.073$	lynx die out; $\to (25, 0)$
B	60	E^* stable focus	$-0.188 \pm 0.515i$	$\to$ (30, 11.25), e-fold ≈ 5 yr
C	85	just below K_H	$-0.022 \pm 0.623i$	$\to$ (30, 14.56), e-fold ≈ 45 yr
D	90	Hopf point	$\pm 0.632i$	very slow (algebraic) decay
E	95	just above K_H	$+0.020 \pm 0.640i$	cycle: $H \in [15.8, 50.9]$, $P \in [9.8, 20.8]$, $T = 10.25$
F	150	well above K_H	$+0.150 \pm 0.676i$	cycle: $H \in [1.1, 134.3]$, $P \in [1.7, 41.2]$, $T = 13.91$

Two cautions for the numerics. At D the equilibrium is a weak focus ($\ell_1 < 0$), so oscillations decay only like $t^{-1/2}$; after 3000 years the residual amplitude in H is still about 1.5, which is expected and is not a sign of a cycle. The cycle at E is already large because the amplitude coefficient is large ($A_H \approx 7.8\sqrt{K - K_H}$), so "small amplitude" only holds within about one unit of K_H.

1.9 Sensitivity of the threshold

Because $K_H = b(ea+m)/(ea-m)$ is explicit, sensitivity can be read off exactly. The elasticities (percentage change in K_H per percentage change in a parameter) are

$$\frac{\partial \ln K_H}{\partial \ln b} = 1, \qquad \frac{\partial \ln K_H}{\partial \ln m} = -\frac{\partial \ln K_H}{\partial \ln a} = -\frac{\partial \ln K_H}{\partial \ln e} = \frac{2\,ea\,m}{(ea)^2 - m^2}, \qquad \frac{\partial \ln K_H}{\partial \ln r} = 0.$$

At the illustrative values ($ea = 2m$) the middle elasticity is $4/3$: a 10% rise in lynx mortality raises the threshold by about 13%, and so stabilises the system. The elasticity blows up as $m \to ea$, where the predator is barely viable and $K_H \to \infty$. Weak predators stabilise.

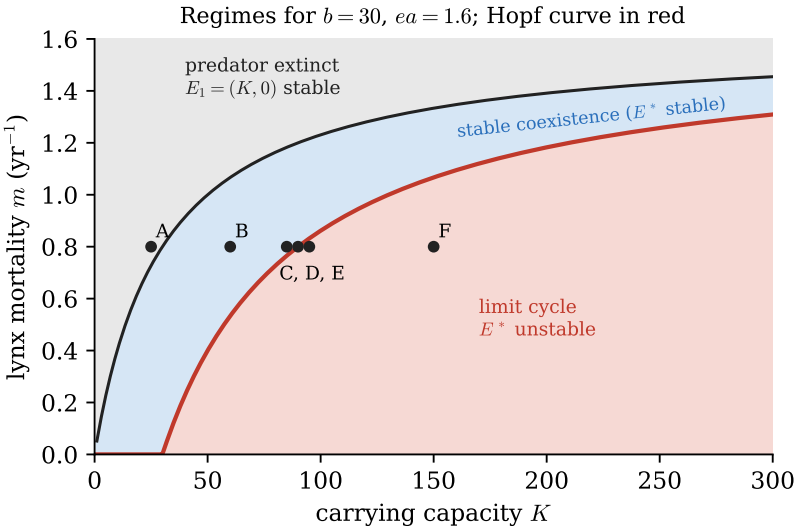


Figure 3: Analytic regimes in the (K, m) plane for $b = 30$, $ea = 1.6$. Black curve: $m = eaK/(b+K)$ (predator persistence boundary). Red curve: $m = ea(K-b)/(K+b)$ (Hopf curve). Dots mark regimes A–F of Table 2.

Equivalently, for fixed K the regimes are bands in m (Figure 3):

$$\underbrace{m < \frac{ea(K-b)}{K+b}}_{\text{limit cycle}} \qquad \underbrace{\frac{ea(K-b)}{K+b} < m < \frac{eaK}{K+b}}_{\text{stable coexistence}} \qquad \underbrace{m > \frac{eaK}{K+b}}_{\text{lynx extinct}}$$

A useful single number for any fitted parameter set is the *stability margin* $\sigma = K/K_H$: $\sigma < 1$ means damped oscillations, $\sigma > 1$ a limit cycle. Reporting σ (with its spread over fits) is the cleanest way to say whether the data support sustained cycling.

Two structural facts matter for inference. First, a and e enter H^*, K_H and the stability conditions only through the product ea. Second, if lynx are measured only up to an unknown multiplicative factor c (as pelt counts are), the map $(P, a, e) \mapsto (cP,\ a/c,\ ec)$ leaves (1) unchanged (checked symbolically), so a and e are *not separately identifiable* from index data. Fitting should fix one of them, or fit ea and a scale, rather than report both as biological rates.

1.10 Dimensionless form of the results

With the scaling $x = H/K$, $y = aP/(rK)$, $s = rt$ (to be reconciled with the frozen scaling in the model section), (1) becomes

$$x' = x(1-x) - \frac{xy}{\beta + x}, \qquad y' = y\Big(\frac{\alpha x}{\beta + x} - \delta\Big), \qquad \beta = \frac{b}{K},\ \alpha = \frac{ea}{r},\ \delta = \frac{m}{r}.$$

Then $x^* = \beta\delta/(\alpha - \delta)$, coexistence requires $\alpha > \delta$ and $x^* < 1$, and the Hopf condition $K = K_H$ reads $\beta(\alpha + \delta)/(\alpha - \delta) = 1$, i.e. $\beta + 2x^* = 1$. Only three combinations of the six parameters control the qualitative dynamics.

Table 3: Status of each stability claim in this section.

Claim	Evidence
Equilibria, feasibility, local stability of E_0, E_1, E^*	Derived here; SymPy-checked
Hopf threshold $K_H = b + 2H^*$, $\Delta > 0$, transversality	Derived here; SymPy-checked
A periodic orbit exists for every $K > K_H$	Derived here (Poincaré–Bendixson)
Cycle is unique and attracting for $K > K_H$	Cited theorem [1]
E^* globally stable for $H^* < K < K_H$	Cited theorem [2]
Supercritical Hopf ($\ell_1 < 0$)	Numerical evaluation of analytic formula, two parameter sets
$\sqrt{K - K_H}$ amplitude law; periods, amplitudes, lags	Numerical (RK45), illustrative parameters
Real hare–lynx data lie on a limit cycle	*Not established here*; requires $\sigma > 1$ at fitted values with uncertainty

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